CS5489
Lecture 1.1: Machine Learning - Overview

Kede Ma

City University of Hong Kong (Dongguan)



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Outline

1 Course Information

2 Introduction

3 Review and Notation

Basic Information

- **Teaching Team**
  - Instructor: Kede MA
    - kede.ma@cityu.edu.hk, Office: AC4-547
  - **Lead TA**: Jiaqi He
    - jiaqi.he@my.cityu.edu.hk
  - **TA**: Shuyan Zhai
  - **TA**: Tianhe Wu
  - **TA**: Rongguang Ye
  - **Postdoc Helper**: Xin Chen
  - **Postdoc Helper**: Zengming Wang
  - **MPhil Helper**: Zhengpeng Xing
- **Canvas-Based Course Site**
  - Students are responsible for checking Canvas and their University email accounts regularly for announcements and updates

Teaching Activities (In-Person)

- **Lecture** (two hours per week)
  - Present machine learning algorithms with emphasis on how those algorithms are **mathematically** derived and **practically** applied
- **Tutorial** (eight in total, schedule TBD)
  - Use machine learning algorithms on small examples to gain better intuition and understanding
- **Homework Assignment** (four in total)
  - Solve mathematical problems in machine learning
- **Course Project** (one)
  - Apply machine learning to solve a real-world problem
  - Up to **five** students per group

Assessment

- **Coursework** (70%)
  - Tutorial exercises (10%) - due one week (after release)
  - Assignments (30%) - due two weeks (after release)
  - Course project (30%) - due Week 13 (tentative)
    - Project report, code implementation, and (optional) presentation
- **Final Exam** (30%)
- **Note**:
  - Must get at least 30% on the final exam and at least 30% on the course project to pass the course

Reference Books

The course recommends several books freely available:

- [IMLP]: *Introduction to Machine Learning with Python*. Andreas C. Müller and Sarah Guido
- [PRML]: *Pattern Recognition and Machine Learning*. Christopher M. Bishop
- [CO]: *Convex Optimization*. Stephen P. Boyd and Lieven Vandenberghe
- [MC]: *The Matrix Cookbook*. Kaare B. Petersen and Michael S. Pedersen
- [LA]: *Linear Algebra and Its Applications*. Gilbert Strang

Course Abstract

- The goal of this course is to introduce students to the field of machine learning
  - Machine learning algorithms allow computers to automatically learn to recognize complex patterns from empirical data, such as text and web documents, sounds, images, and videos
  - This course is intended to give a broad overview of machine learning from both **theoretical** and **practical** standpoints, with emphasis on applying it to solve real-world problems
  - At the end of the course, students will have both mathematical and practical experience with machine learning algorithms

Therefore, on some occasions, you may feel like you are sitting in some **math** class. If you have difficulty keeping up the course, you should drop it during the add-drop period. No late drops are allowed

CILOs

- Identify and explain common machine learning algorithms
- Apply machine learning algorithms to solve real-world problems
- Evaluate the effectiveness of different machine learning algorithms and discuss their advantages and disadvantages
- Describe the theoretical and practical aspects of machine learning algorithms

Prerequisites

The course has formal prerequisites as listed below. All students are expected to be familiar with this material or have the ability to make up any gaps in their backgrounds

- Linear Algebra (e.g., matrix multiplication, linear independence)
- Calculus (e.g., continuity, differentiability)
- Probability and Statistics (e.g., random variable, multivariate Gaussian distribution)
- Optimization (e.g., convexity, duality)

This course requires the use of Python for programming. Students are expected to learn Python as we go

Self-Evaluation: Am I Ready to Go?

- What is the derivative of $\mathbf{x}^T\mathbf{w}$, w.r.t. $\mathbf{w}$, where $\mathbf{x}^T \in \mathbb{R}^{1\times N}$ is a row vector and $\mathbf{w} \in \mathbb{R}^{N\times 1}$ is a column vector?
- What is maximum likelihood estimation (MLE)? What is maximum a posteriori (MAP) estimation?
- What is gradient descent? What is stochastic gradient descent?
- What is principal component analysis (PCA)? What is singular value decomposition (SVD)?

3-4: Good to go; **0-2**: Don't worry - we'll help you catch up

Academic Honesty

- CityUHK (Dongguan) has its own **Rules of Academic Honesty**
- Plagiarism...
  - It is serious fraud to plagiarize others' work
  - Punishment ranges from warning to course failure
- How to prevent plagiarism...
  - Finish the assignments by yourself! You have to write the program/solution yourself
    - Okay to talk about how to do the problem with your classmates
    - Protect your solution; don't give it away as a "reference" copy
  - In plagiarism cases, we treat both giver and copier as guilty
  - You hurt your own grades by not reporting cheating
- As instructors...
  - We have the responsibility to report academic dishonesty cases so as not to compromise the quality of education
  - We take suspected plagiarism cases very seriously

Academic Honesty in the Presence of GenAI

- We basically follow the department's policy on using GenAI (e.g., Codex, Claude, Deepseek, and Kimi)
  - For **Summative Assessment** like Quiz/Assignment/Pre-mid-term Test/Midterm Exam/Final Exam, students are generally NOT permitted to use GenAI tools
  - For any other **Formative Assessment** like coding and writing, students, by default, should be permitted to use GenAI tools
- We will introduce the underlying mechanisms of some popular generative models towards the end of this course

Outline

1 Course Information

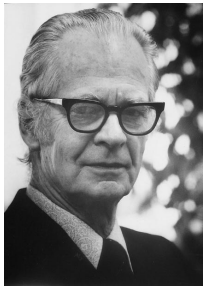
2 Introduction

3 Review and Notation

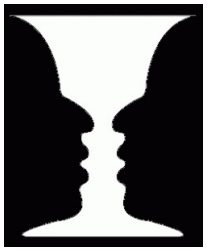
Introduction

What is Learning?

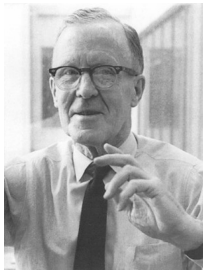
Definitions of Learning



Behaviorism (Skinner, 1900-1950): Learning is a long-term change in behavior due to experience



Cognitivism (Gestalt School, 1920-): Learning is an internal mental process that integrates new information into established mental frameworks and updates those frameworks over time



Connectionism (Hebb, 1949): Learning is a physical process in which neurons join by developing the synapses between them

Introduction

What is Machine Learning?

Views on Machine Learning



Samuel (1959): "Machine learning is a field of study that gives computers the ability to learn without being explicitly programmed"



Mitchell (1997): "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E"



Jordan (2015): "It is one of today's rapidly growing technical fields, lying at the intersection of computer science and statistics, and at the core of artificial intelligence and data science"

Samuel's View

- Example: Recognizing handwritten digits in an image
  - 28×28 image $\longrightarrow$ 784-dim vector
  - A lot of variations in handwriting, position, scale, and stroke
  - Difficult to identify rules & code by hand



- ML solution:
  - Gather some example data
  - Train computer to discover differences automatically

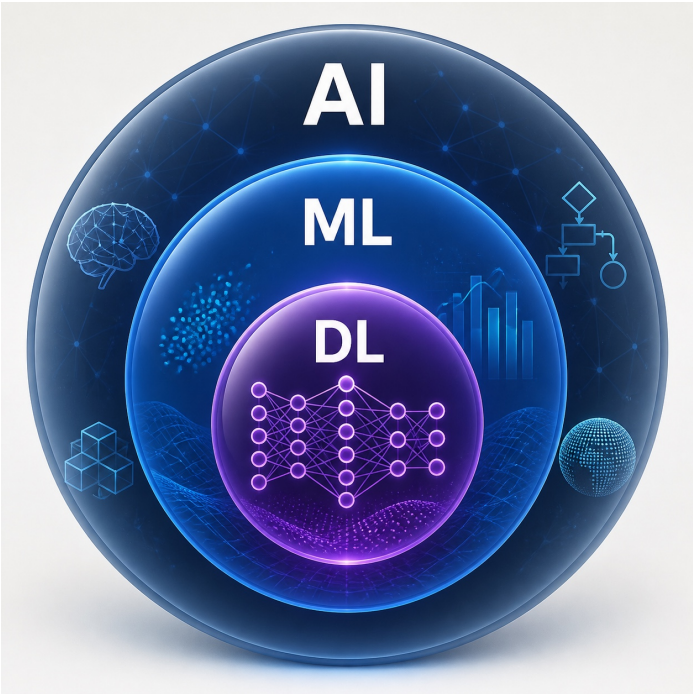
Mitchell's View

- "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E"
  - "class of tasks T": Learning is task-specific (recognition, clustering, etc.)
  - "performance measure P": Optimize a loss function (e.g., error rate), but also prevent overfitting (regularization)
  - "experience E": Data-driven! More data is better!

Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL)

How to Read the Diagram

- AI is the broad discipline
- ML is one family of approaches within AI
- Deep learning is one family within ML
- Vision, language, and robotics are application areas that may use ML



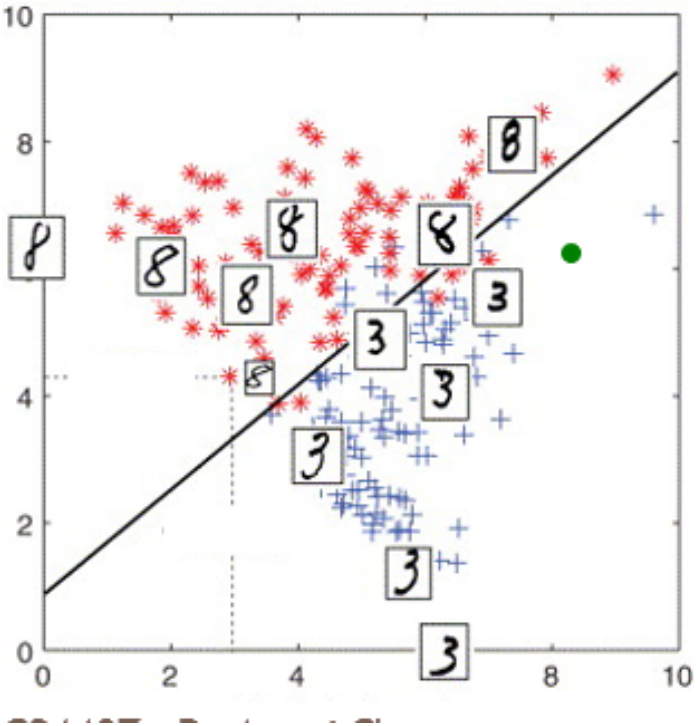
Topics in Machine Learning

- Supervised learning
- Unsupervised learning
- Reinforcement learning
- Learning theory

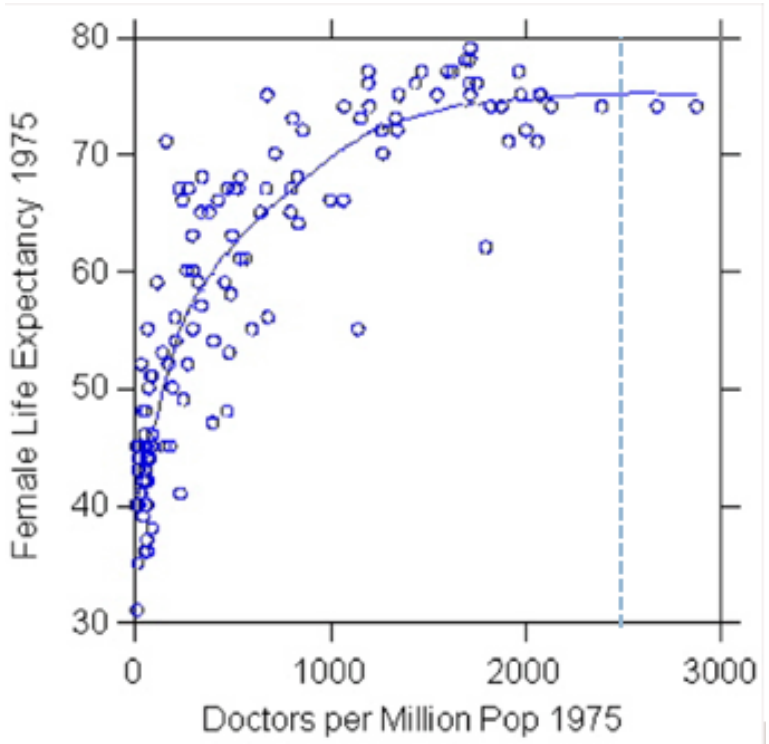
Supervised Learning

- Training data has inputs and outputs
  - E.g., digit recognition (input=image, output=digit)
  - Learn a function mapping inputs to outputs

Classification

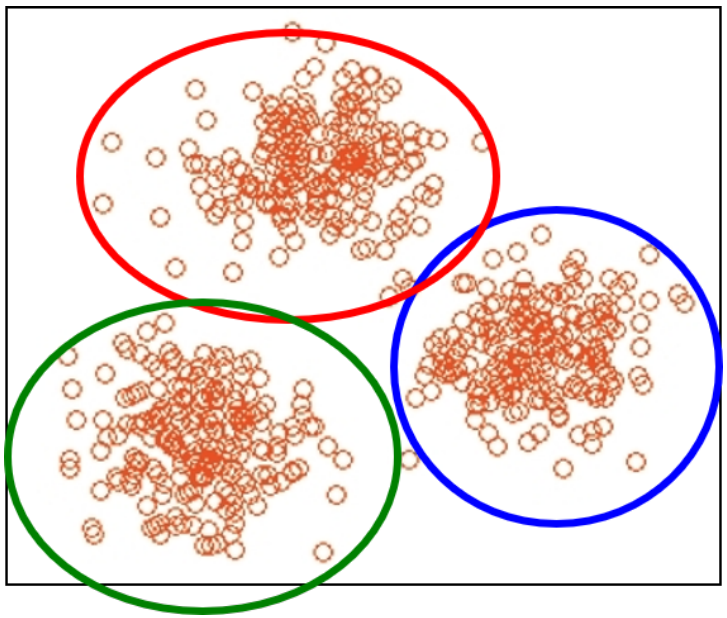


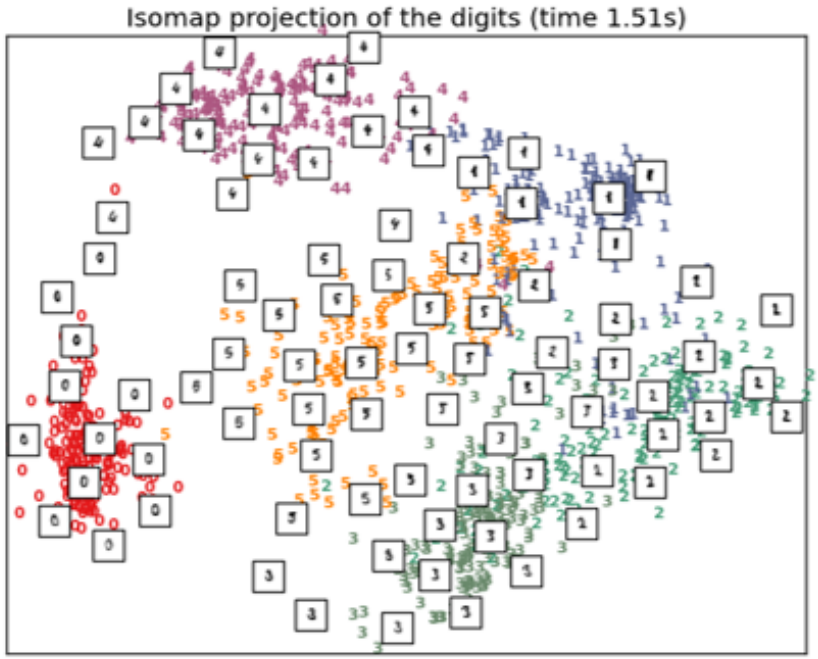
Regression



Unsupervised Learning

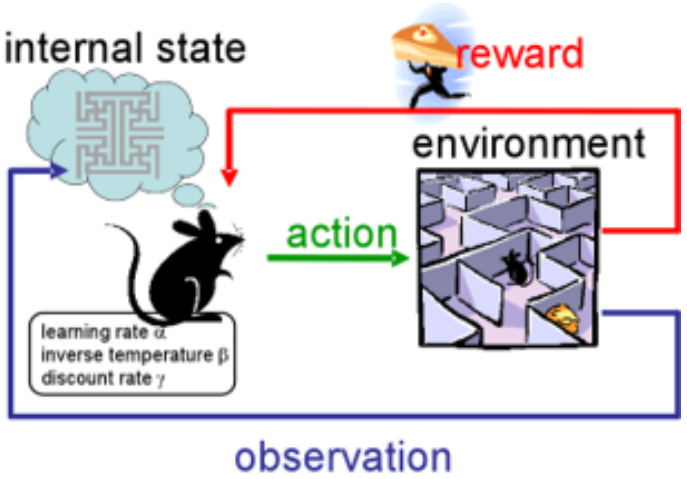
- Training data only has inputs (no outputs)
  - Density estimation - construct a probability model over the input
  - Clustering - discover groups of similar examples
  - Dimensionality reduction - project high-dim data to 2 or 3-dimensions (for visualization purposes)





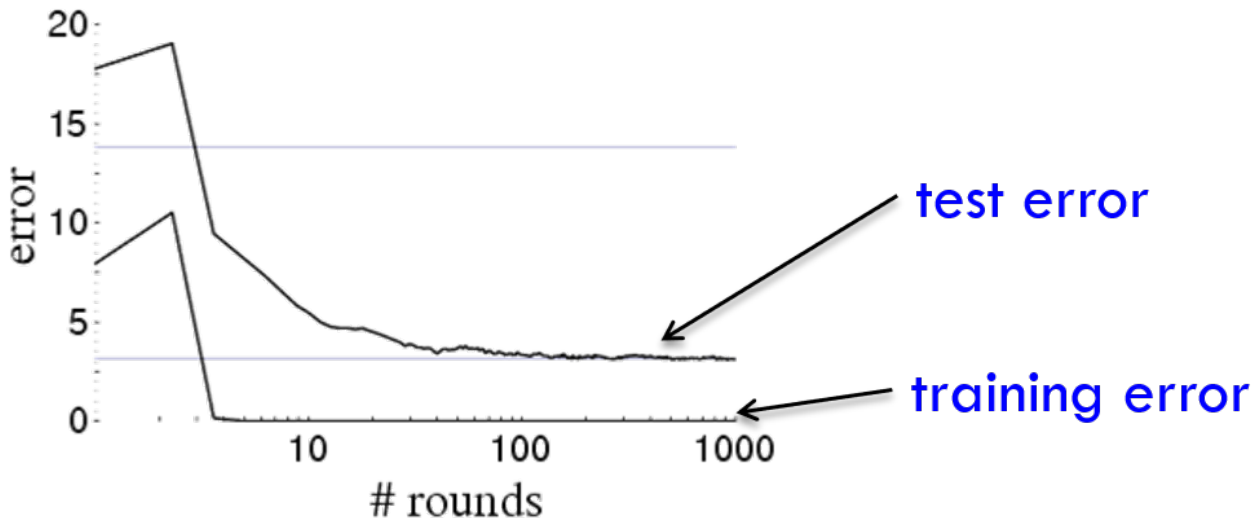
Reinforcement Learning

- Make a sequence of actions, given current states
  - E.g. a robot interacting with its environment
  - Maximize the reward
    - At some point, receive a reward or a punishment
    - Actions may also affect future reward



Learning Theory

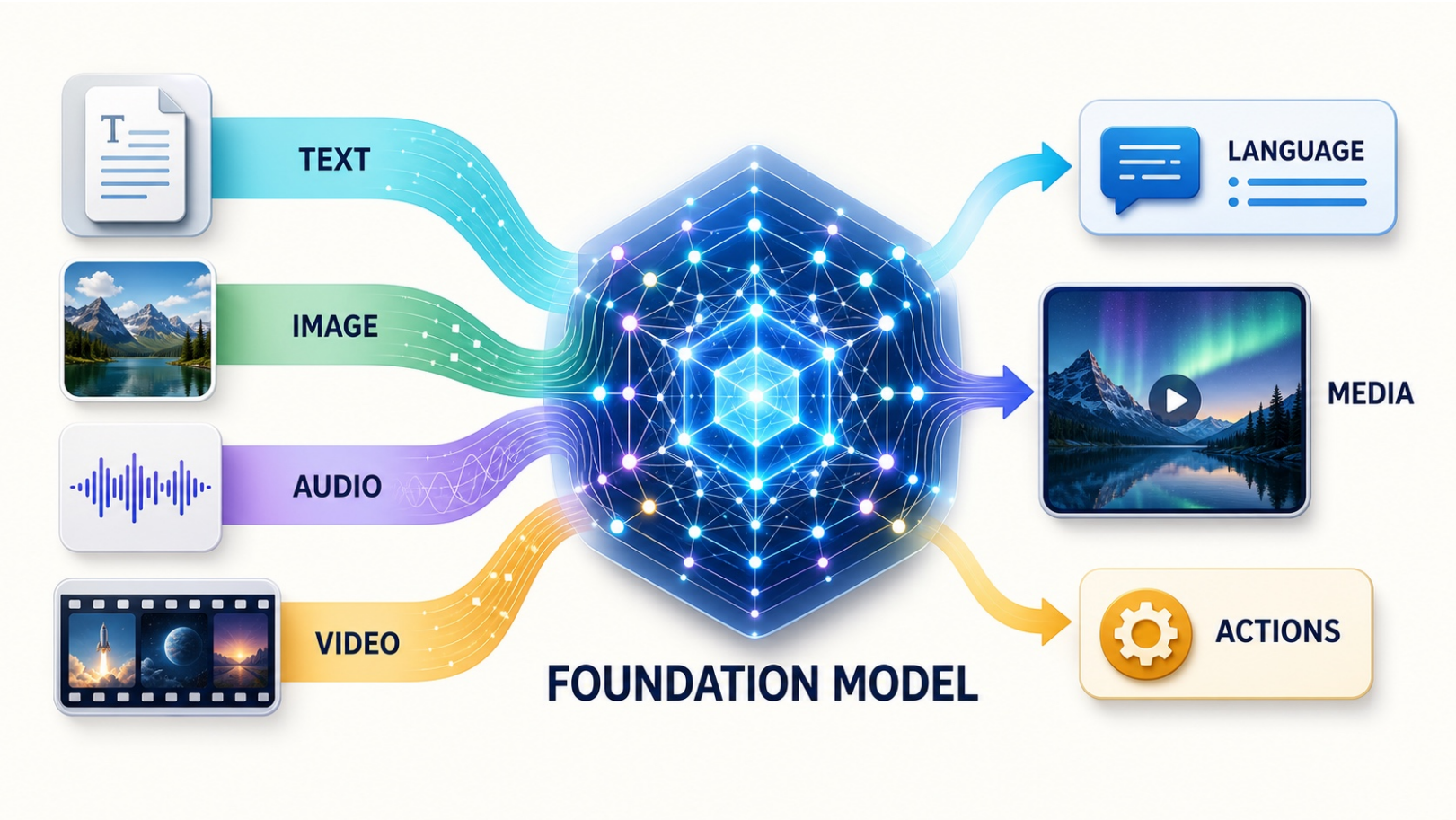
- Why does machine learning work?
  - Performance guarantees - bounds on the expected test error
  - What types of functions can be represented by an algorithm, and how much data do we need?



Machine Learning in Practice



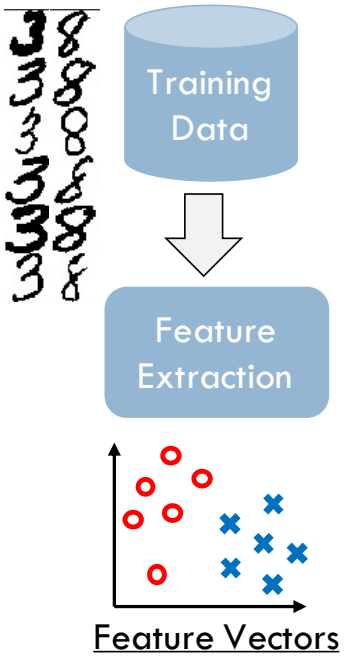
Generative AI: Multimodal Foundation Models



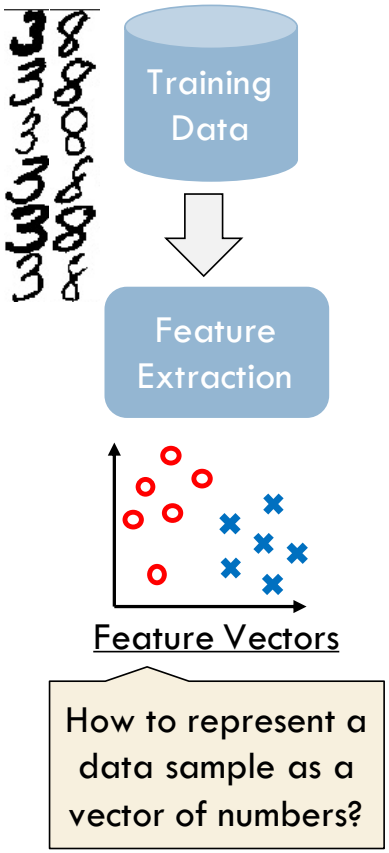
Machine Learning Training Pipeline



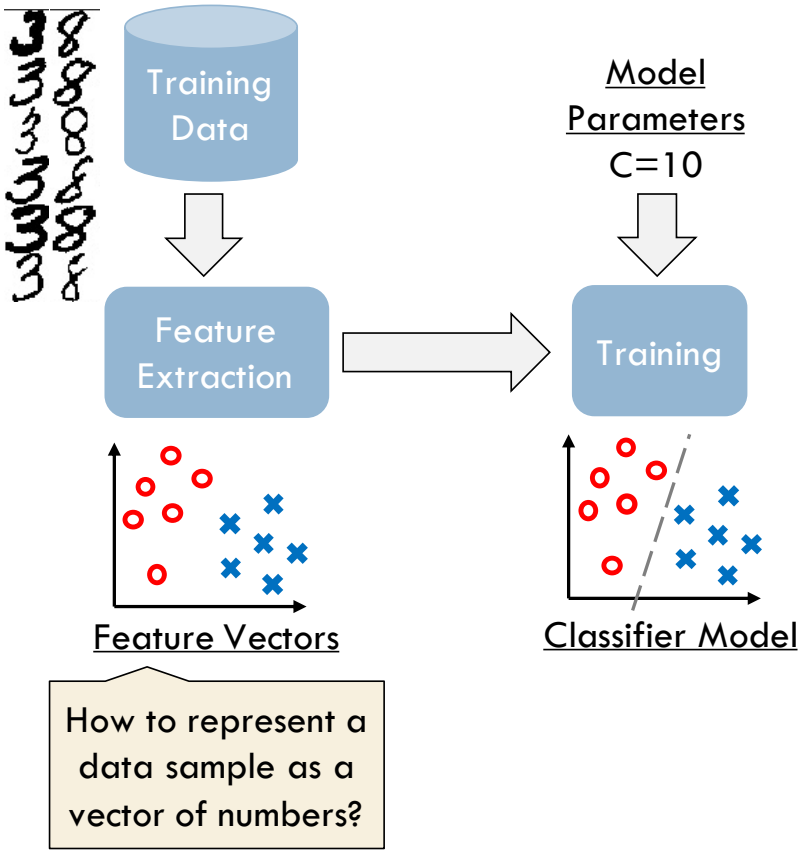
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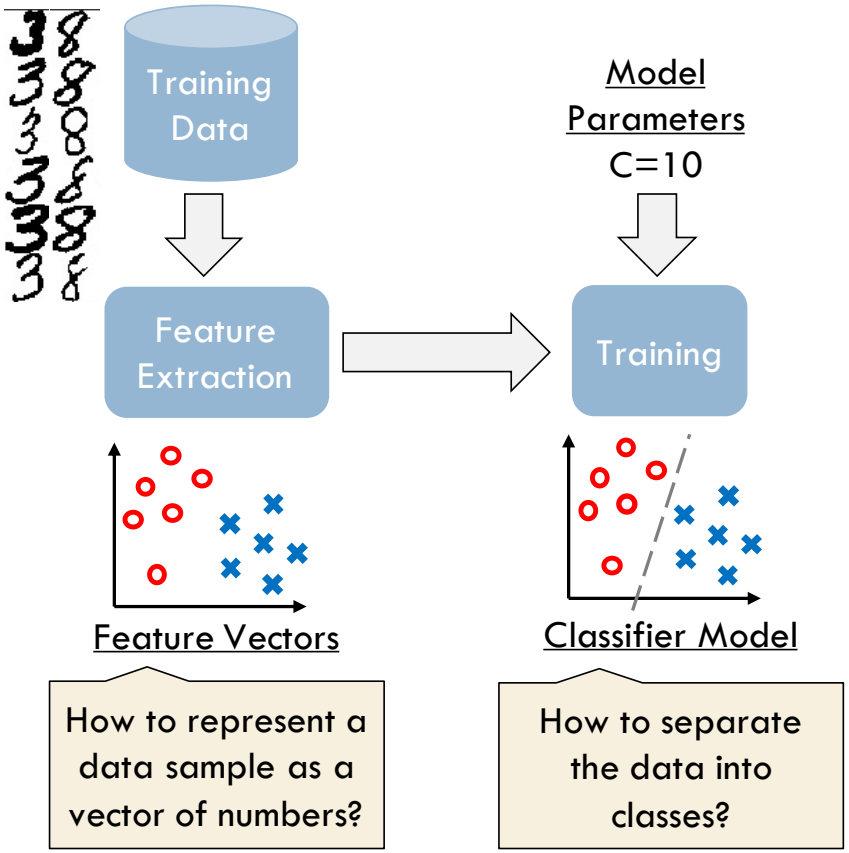
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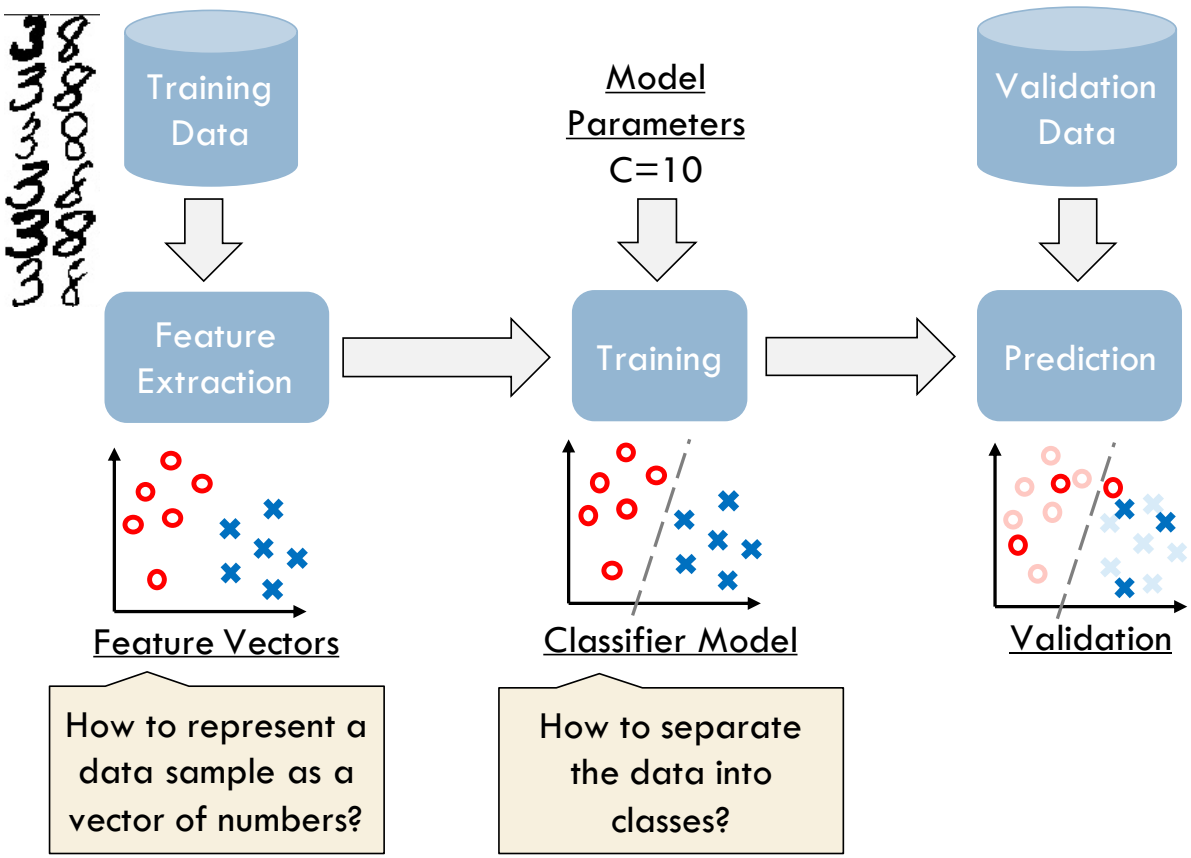
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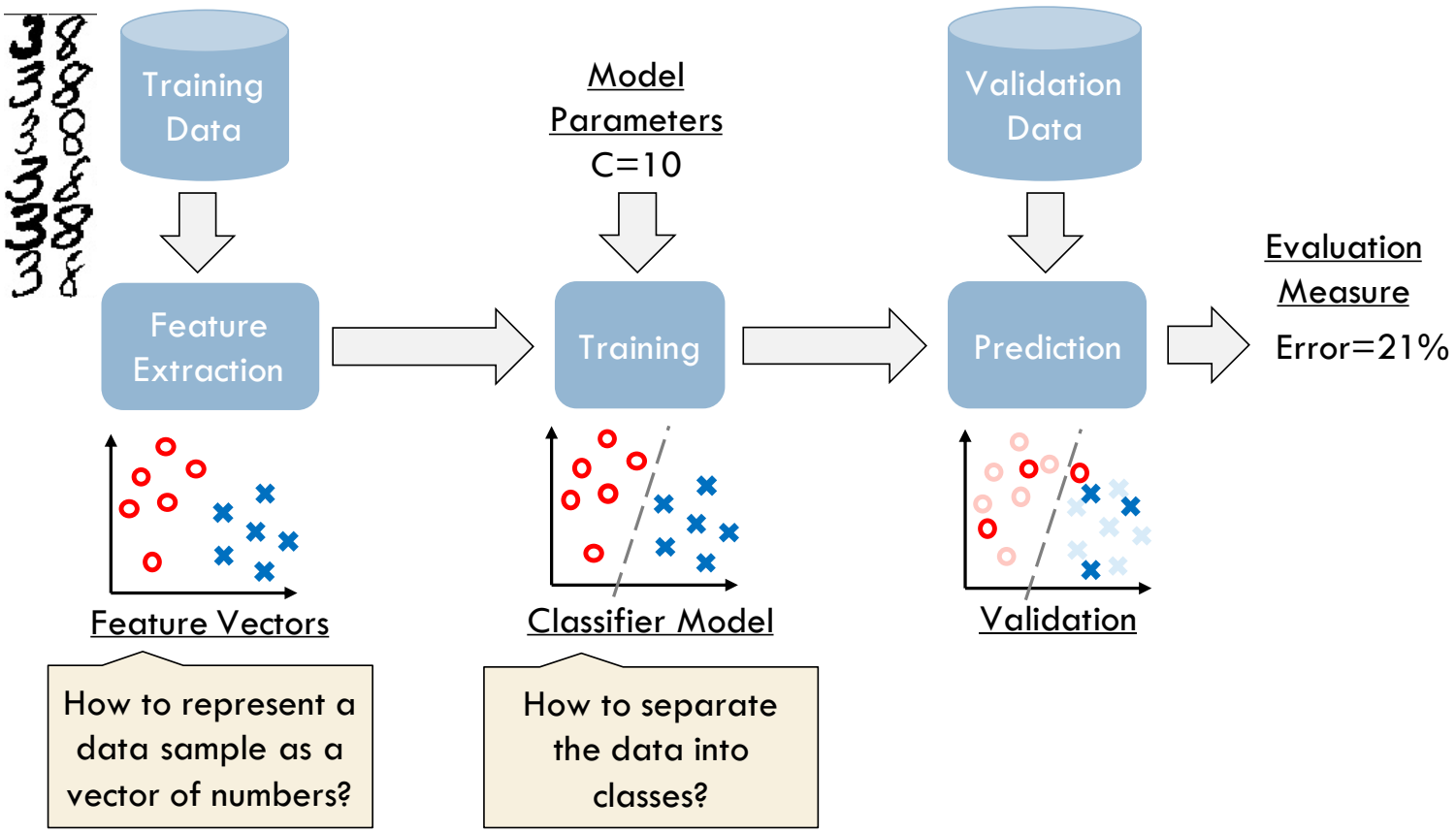
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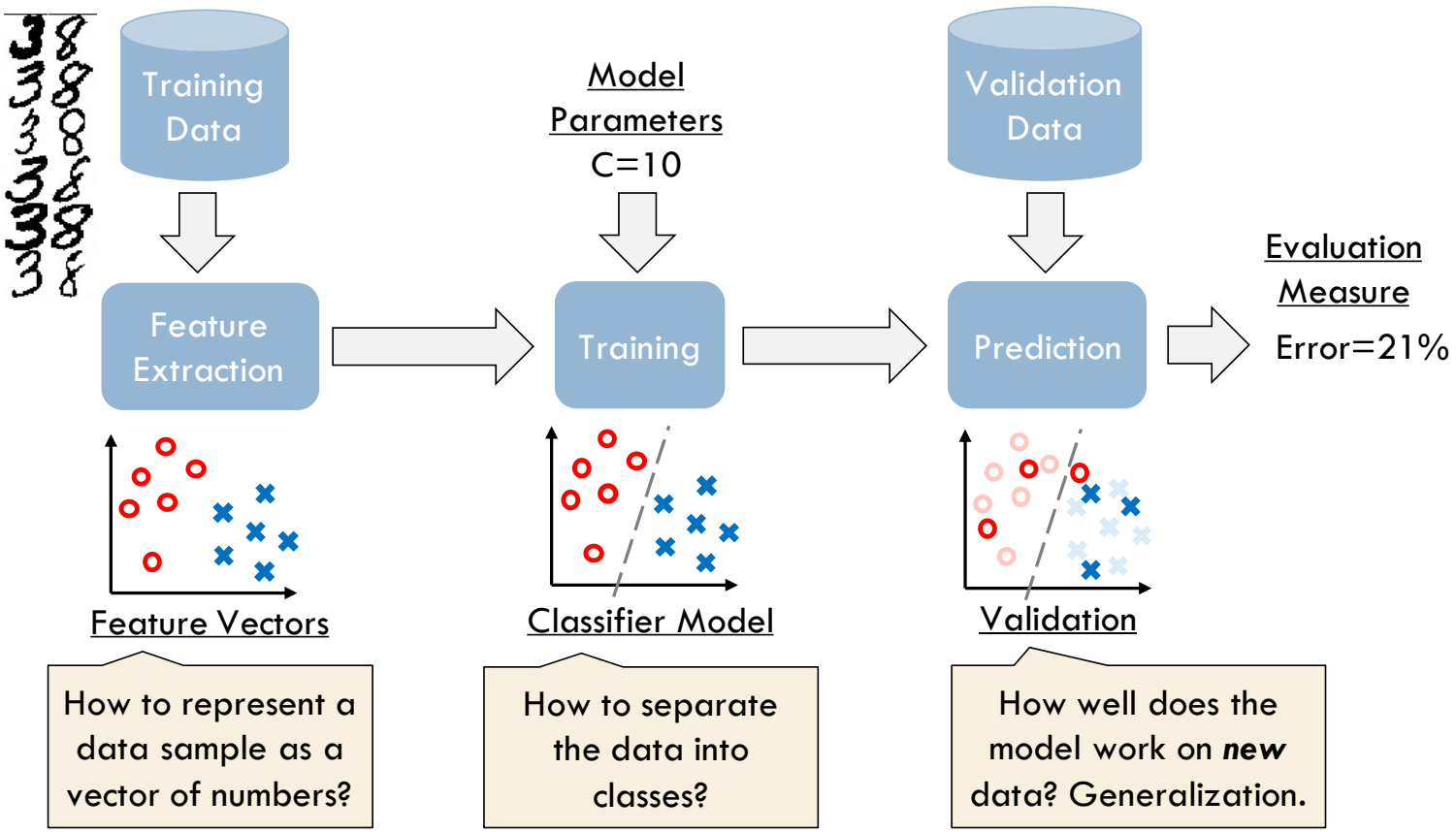
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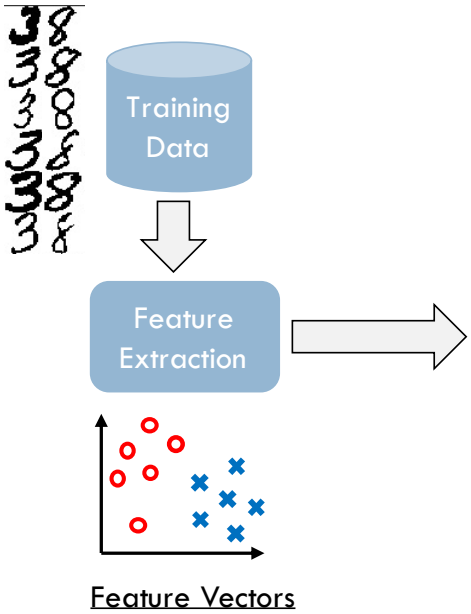
Machine Learning Training Pipeline



Machine Learning Training Pipeline

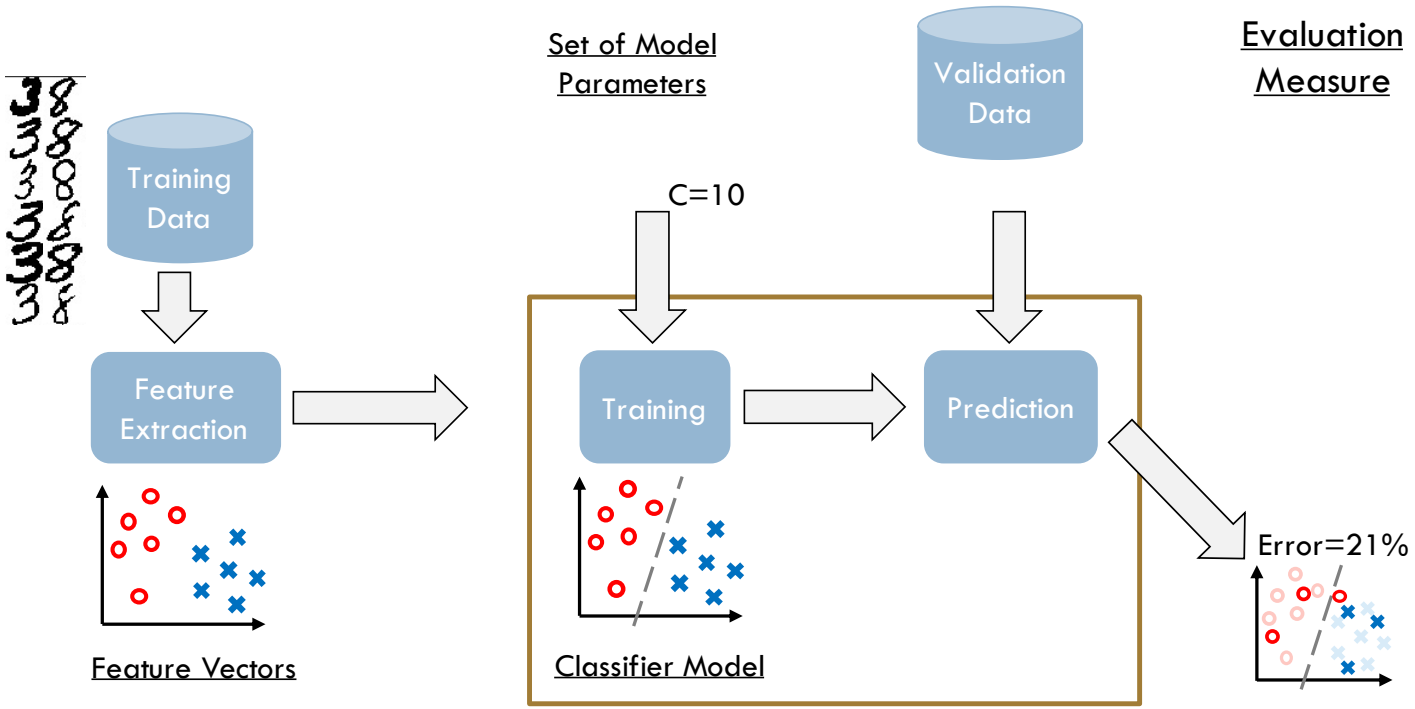


Model Selection Pipeline

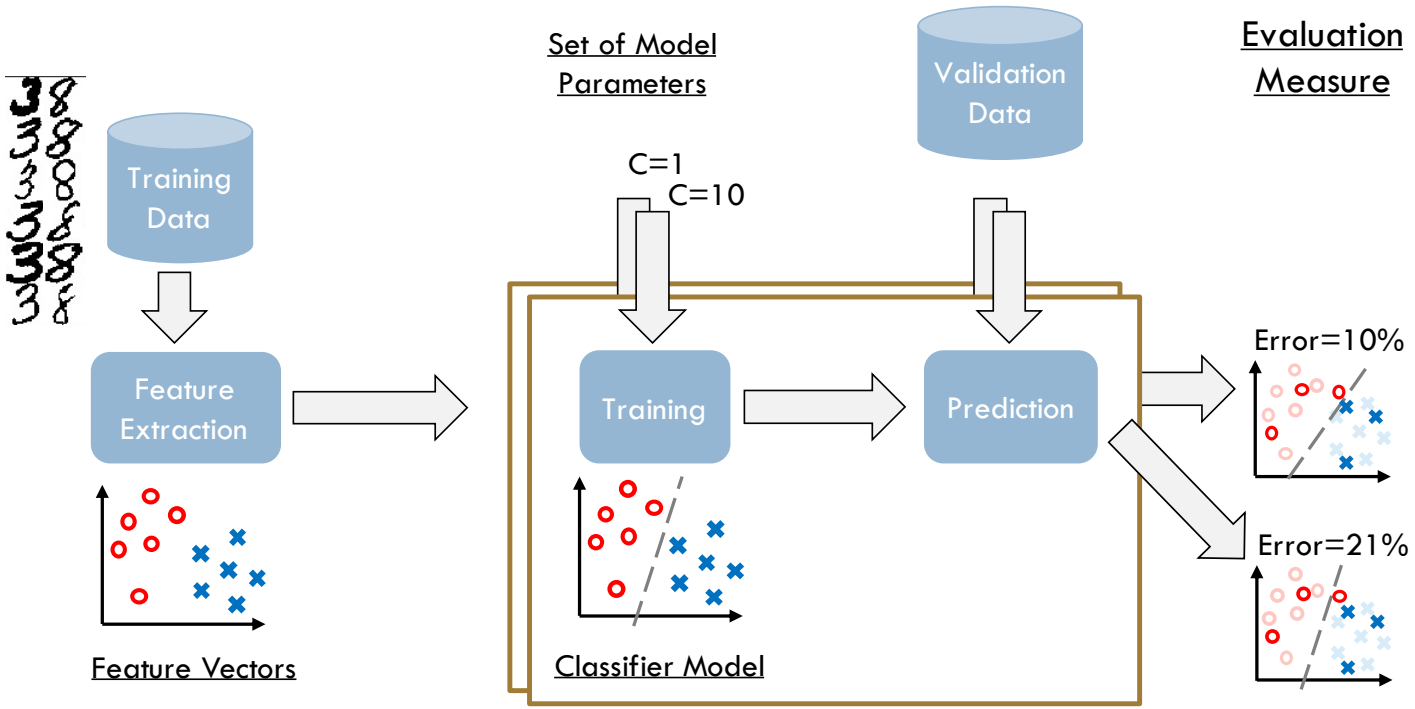


Evaluation Measure

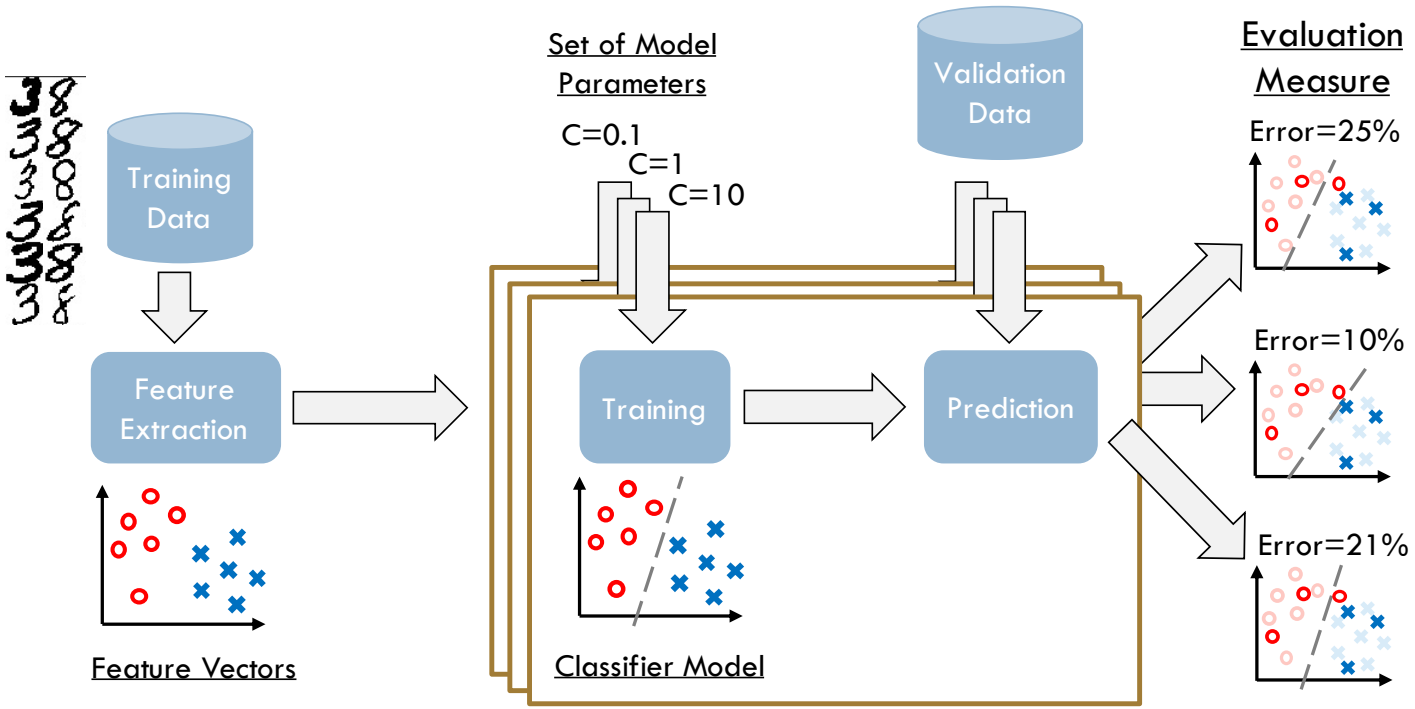
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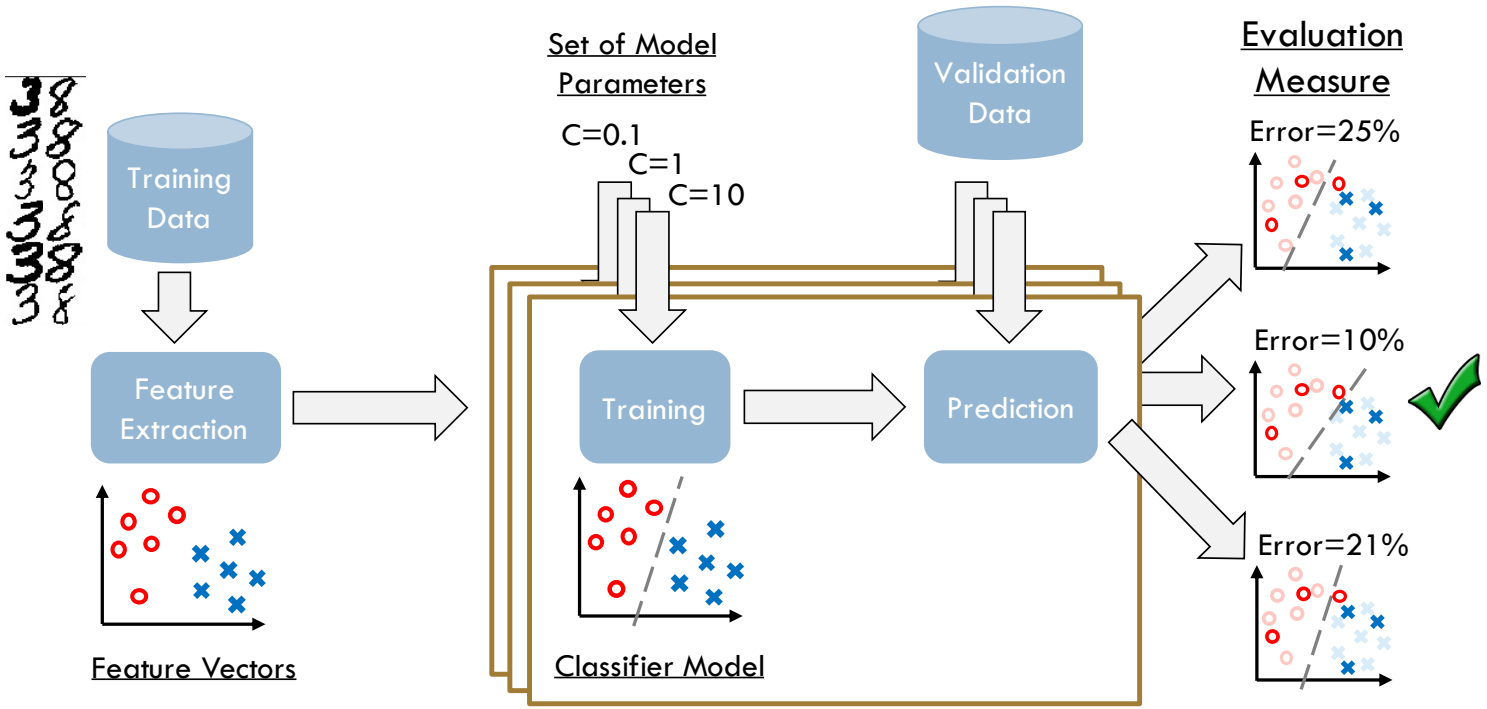
Model Selection Pipeline



Model Selection Pipeline



Model Selection Pipeline



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Linear Algebra

Definition: Vector Space

The real vector space $\mathbb{R}^N$ is a set with elements $\mathbf{x} = [x_1, ..., x_N]^T$ where each $x_j \in \mathbb{R}$. The elements $\mathbf{x}$ are called vectors, and they satisfy the following properties:

- **Addition:** If $\mathbf{x} \in \mathbb{R}^N$ and $\mathbf{y} \in \mathbb{R}^N$, then $\mathbf{x} + \mathbf{y} = [x_1 + y_1, ..., x_N + y_N]^T \in \mathbb{R}^N$
- **Scalar Product:** If $\mathbf{x} \in \mathbb{R}^N$ and $a \in \mathbb{R}$, then $a\mathbf{x} = [ax_1, ..., ax_N]^T \in \mathbb{R}^N$
- **Inner Product:** If $\mathbf{x} \in \mathbb{R}^N$ and $\mathbf{y} \in \mathbb{R}^N$, then $\mathbf{x}^T\mathbf{y} = \sum_{j=1}^{N} x_j \cdot y_j$

Linear Algebra

Definition: Matrix

A matrix $\mathbf{X} \in \mathbb{R}^{M \times N}$ is a rectangular array of elements $x_{ij} \in \mathbb{R}$, $1 \leq i \leq M$, $1 \leq j \leq N$:

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1N} \\ x_{21} & x_{22} & \cdots & x_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ x_{M1} & x_{M2} & \cdots & x_{MN} \end{pmatrix}$$

Linear Algebra

Definition: Matrix

A matrix $\mathbf{X} \in \mathbb{R}^{M \times N}$ supports the following operations:

- **Addition:** If $\mathbf{X} \in \mathbb{R}^{M \times N}$, $\mathbf{Y} \in \mathbb{R}^{M \times N}$ and $\mathbf{Z} = \mathbf{X} + \mathbf{Y}$, then $\mathbf{Z} \in \mathbb{R}^{M \times N}$ and $z_{ij} = x_{ij} + y_{ij}$
- **Scalar Product:** If $\mathbf{X} \in \mathbb{R}^{M \times N}$, $a \in \mathbb{R}$ and $\mathbf{Z} = a\mathbf{X}$, then $\mathbf{Z} \in \mathbb{R}^{M \times N}$ and $z_{ij} = ax_{ij}$
- **Matrix Multiplication:** If $\mathbf{X} \in \mathbb{R}^{M \times N}$, $\mathbf{Y} \in \mathbb{R}^{N \times C}$ and $\mathbf{Z} = \mathbf{XY}$, then $\mathbf{Z} \in \mathbb{R}^{M \times C}$ and $z_{ij} = \sum_{k=1}^{N} x_{ik} y_{kj}$

You should be familiar with basic matrix types (square, diagonal, identity), basic matrix operations (transpose, inverse, trace, etc.), and matrix concepts (eigenvalues, orthogonality, etc.)

Probability

Definition: Probability Distribution

A probability distribution p over a sample space Ω is a **function** from elements/subsets of Ω to real numbers that satisfies the following conditions:

- Non-negativity: $p(\omega) \geq 0$ for all $\omega \subseteq \Omega$
- Normalization: $p(\Omega) = 1$
- Additivity: For all $\omega, \omega' \subseteq \Omega$ that are **disjoint** sets, $p(\omega \bigcup \omega') = p(\omega) + p(\omega')$

Random Variable

Definition: Random Variable

A random variable X is defined by a **function** f_X that maps each element ω of the sample space Ω to a value $x = f_X(\omega)$ in a set $\mathcal{X}$ (called the *range* of the random variable)

For each $x \in \mathcal{X}$ the **event** $\{X = x\}$ refers to the subset of the sample space $\{\omega | \omega \in \Omega, f_X(\omega) = x\}$

For each $x \in \mathcal{X}$ the probability $p(X = x) = p(\{\omega | \omega \in \Omega, f_X(\omega) = x\})$

Probability and Random Variable

We can also specify a probability distribution for a random variable X with range $\mathcal{X}$ directly instead of via an underlying sample space Ω. The following conditions must hold:

- **Discrete probability mass function:**
  $p(X = x) \geq 0 \ \ \forall x \in \mathcal{X}$ and $\sum_{x \in \mathcal{X}} p(X = x) = 1$
- **Continuous probability density function:**
  $p(X = x) \geq 0 \ \ \forall x \in \mathcal{X}$ and $\int_{\mathcal{X}} p(X = x)dx = 1$

Important Probability Concepts

You should be familiar with the following fundamental concepts from probability theory

- Marginalization
- Conditioning
- Bayes' rule
- Expectation
- Classical distributions (Bernoulli, Multinomial, Gaussian)